Automatic Determination of the Indispensability Index of a Term Within a Specialized Domain

Wassim Chenouf, Ange Colletto, Guillaume Andrea Desaphy, Brice ZOZIME
Supervisors: Olfa MECHI, Nadia YASSINE-DIAB

University of Toulouse
https://www.univ-tlse3.fr/

Abstract. Identifying words that are essential to the understanding of a specialized domain represents an important challenge in information retrieval, education, and knowledge organization. In this paper, we study the notion of term indispensability within a specialized domain, which is the ability to serve as an entry point for understanding a corpus. This problem, although central for these applications, remains difficult to formalize due to its inherently subjective nature and the absence of explicit ground truth. We propose a structured review of the main approaches for estimating term importance within a corpus, grouping them into statistical methods, embedding-based methods, graph-based methods, supervised learning approaches, and hybrid methods. We analyze the underlying principles of each family of methods as well as the types of information they capture, such as frequency, specificity, semantic similarity, or structural centrality. Our analysis shows that no single method is sufficient to robustly model indispensability. Statistical methods are simple but limited in semantic understanding, embedding models capture meaning but may lack interpretability, graph-based approaches reveal structural roles but depend heavily on graph construction, and supervised approaches are constrained by the scarcity of reliable annotations. Overall, hybrid strategies appear most promising by combining complementary signals. Finally, we discuss future directions, particularly the use of advanced language models and evaluation frameworks better suited to this inherently subjective notion.

Keywords: Term Importance · Term Weighting · Keyword Extraction · Natural Language Processing

1 Introduction

Not all terms contribute equally to the understanding of a domain. Some play a central role and serve as entry points to a broader set of concepts, while others are more peripheral, and being able to automatically make this distinction is actually more useful than it might seem. In contexts such as domain-specific language learning, information retrieval, or knowledge organization, knowing which terms to prioritize can make a real difference, whether it helps learners

focus on the most essential vocabulary first, or improves how documents are indexed and queried.

A concrete example is education: if a student is learning a new specialized domain, being able to automatically rank terms by importance would allow them to encounter the most fundamental concepts first, before moving on to more peripheral ones. Rather than going through vocabulary in an arbitrary order, they would build their understanding progressively, starting with the terms around which most other concepts are structured. This is what we refer to as the concept of indispensability of a term within a domain. We therefore aim to compute a score related to this concept. This indispensability score provides a quantitative measure of a term's importance within this specific domain. However, modeling indispensability is particularly challenging. First, the notion itself is inherently subjective: different experts may disagree on the importance of a given term. Second, there is no available ground truth dataset that directly captures this notion. Finally, the estimation of term importance depends heavily on the corpus used to represent the domain, making data quality and domain specificity critical factors.

The construction of the corpus is a critical stage, as it directly shapes how the domain is represented. Since most approaches in this work are unsupervised, they rely entirely on the available data to identify salient concepts. As a result, the quality and characteristics of the corpus strongly influence both the extracted terms and their associated importance scores. An effective corpus should be representative of the target domain and coherent with it. A highly specialized corpus highlights domain-specific terminology, whereas a broader one may introduce noise and promote less relevant terms. Corpus size is also an important factor: a corpus that is too small may lead to unstable or biased estimates, while an excessively large one may dilute domain-specific signals through heterogeneous contexts. In this context, large-scale academic databases such as OpenAlex are particularly useful for constructing corpora, as they provide access to extensive collections of publications along with structured metadata (e.g., keywords, citations, and disciplinary classifications). These features facilitate the selection of relevant and coherent document sets while helping control corpus size and diversity. Overall, constructing an appropriate corpus requires balancing domain specificity, thematic coherence, and sufficient data volume, as any bias or noise can directly affect the reliability of term indispensability estimates.

One of the most intuitive ways to estimate the importance of a term in a corpus is to rely on its frequency. A term that appears frequently in a document or in a domain may naturally be considered important. This idea forms the basis of early approaches in automatic text analysis and information retrieval. However, frequency alone quickly shows its limitations. Very common terms may appear often without carrying meaningful information, while rare but highly informative terms may be overlooked. This issue was studied early in the literature, notably by Salton et al. [9], who introduced the idea that term importance should reflect its ability to discriminate between documents. In this perspective, the most

useful terms are those that maximize the separation between documents within a collection. To address these limitations, a wide range of methods have been proposed.

The objective of this report is to study how our definition of a term's indispensability within a domain can be determined automatically in a text corpus by reviewing various existing approaches. The goal is to understand how these approaches work, what they capture, and what their limitations are. For each family, we discuss what kind of information they capture, whether that is frequency, specificity, semantic similarity, or structural centrality, and how these signals can contribute to estimating how indispensable a term is within a specialized corpus. We also point out the main limitations of these approaches, particularly their dependence on corpus quality, their difficulty with semantic ambiguity, and the fact that there is no stable ground truth for a notion that is inherently subjective. What this review makes clear is that no single method is enough to model indispensability on its own. The literature consistently points in the same direction: combining complementary statistical, semantic and structural signals is the most promising way to approximate an indispensability score in specialized domains.

The remainder of this paper is organized into several sections. Different methods for term extraction and prioritization are presented in Section 2. Section 3 discusses the limitations of existing approaches, while Section 4 outlines future directions for keyword extraction. Finally, Section 5 presents the conclusion.

2 Term extraction and prioritization methods

The computation of a term's indispensability score can be carried out using statistical, NLP, and artificial intelligence methods that were not originally designed for this purpose. The methods we will discuss are generally used for keyword extraction, semantic similarity computation, or evaluating the relevance of a term within a corpus. In this section, we first present term extraction methods. We then analyze term prioritization methods, which we categorize into the following approaches: statistical methods, embedding-based methods, graph-based methods, supervised methods, and hybrid methods.

2.1 Term extraction methods

Term extraction methods aim to identify the right candidates before any scoring is applied by focusing on extracting high-quality phrases that correspond to coherent semantic units. This step really matters when it comes to indispensability, because if the candidates are not well defined to begin with, then whatever scoring method comes after will not be very reliable either. By restricting the analysis to well-formed and domain-relevant expressions, term extraction reduces noise and ensures that importance scores are assigned to meaningful concepts rather than pieces of text that do not really represent anything.

Classical term extraction approaches rely on candidate generation based on simple patterns, such as frequent n-grams or noun phrases identified through syntactic patterns [13]. However, these methods often produce a large number of noisy candidates, including incomplete or irrelevant expressions.

More recent methods, such as AutoPhrase [10], propose a more robust approach by introducing the notion of *quality phrases*. The process relies on several steps: extraction of frequent candidates, estimation of their quality using external corpora or statistical signals, and then text segmentation to identify the most coherent expressions. This approach makes it possible to effectively filter out incorrect or uninformative expressions while remaining adaptable to different domains.

2.2 Statistical methods

Statistical methods capture term importance through distributional properties within the corpus. This allows them to identify terms that are both locally representative and globally discriminative, a principle already highlighted by Salton et al. [9]. More advanced statistical methods, such as YAKE or RAKE, extend this idea by adding additional signals including term position, dispersion, and co-occurrence patterns.

TF-IDF: A first response to the limitations of raw frequency is TF-IDF, one of the most well-known methods in natural language processing. The idea is quite intuitive: a term is important if it appears frequently in a document, but not if it appears everywhere in the corpus. We therefore want to retain terms that are locally frequent while penalizing those that are too generic.

Formally, for a term t in a document d belonging to a corpus D, the TF-IDF score is defined as follows:

$$\text{TF-IDF}(t, d, D) = \text{TF}(t, d) \times \text{IDF}(t, D) \tag{1}$$

with

$$\text{IDF}(t, D) = \log \frac{|D|}{|\{d' \in D : t \in d'\}|} \tag{2}$$

In practice: if a term appears frequently in a single document (high TF) but rarely in others (high IDF), its score will be high. Conversely, a word such as *"the"* or *"is"* may be very frequent, but its IDF close to zero will lower its overall score. The TF component measures how often the term appears in the document, while the IDF decreases when the term is present in many documents of the corpus. As a result, terms that are locally frequent but globally rare obtain the highest scores.

This logic is consistent with the work of Salton et al. [9], who already showed that the importance of a term should reflect its ability to discriminate between

documents, and not merely its raw frequency. Ushio et al. [12] confirm that TF-IDF remains a strong baseline, while showing that measures such as lexical specificity can outperform it in certain datasets.

From the perspective of indispensability, TF-IDF provides a useful first approximation. However, it remains limited: it treats terms independently, ignores synonymy, and does not take into account the structural role of a term within a domain. This motivates us to explore richer approaches.

TF-SYN-NER-Rel: Given the limitations of the TF-IDF method, Rodionov et al. [7] propose a new word weighting method designed to complement the classical TF-IDF. This method is called TF-SYN-NER-Rel. It aims to improve the coherence and interpretability of topic modeling approaches based on LDA (Latent Dirichlet Allocation).

The proposed method enriches the weighting of terms by integrating several linguistic dimensions: the position of the word in the document, its syntactic role, its membership in a factual core (based on subject–verb–object relations), as well as its belonging to a named entity (persons, organizations, locations). Each occurrence of a term t_i is assigned an individual weight w_i calculated as the product of four coefficients: a positional coefficient (λ_{pos}), a syntactic coefficient (λ_{syn}), a factual coefficient (λ_{fact}), and a named-entity-based coefficient (λ_{ner}). This leads to the following formula:

$$w_i = \lambda_{\text{pos}}(t_i) \times \lambda_{\text{syn}}(ti) \times \lambda_{\text{fact}}(ti) \times \lambda_{\text{ner}}(ti) \tag{3}$$

with

$$\lambda_{\text{pos}}(t_i) = \begin{cases} 2.0, & \text{if } t_i \text{ placed in header;} \\ 1.5, & \text{if } t_i \text{ placed in 1st sentence;} \\ 1.0, & \text{in other cases.} \end{cases} \tag{4}$$

$$\lambda_{\text{syn}}(t_i) = \begin{cases} 2.0, & \text{if } t_i \text{ is 'root' (predicate, action core);} \\ 1.8, & \text{if } t_i \text{ is 'nsubj' (subject, actor);} \\ 1.5, & \text{if } t_i \text{ is 'obj' (direct object, action target);} \\ 1.1, & \text{if } t_i \text{ is 'amod' (attribute refining the actor);} \\ 1.0, & \text{for all other dependency relations.} \end{cases} \tag{5}$$

$$\lambda_{\text{fact}}(t_i) = \begin{cases} 1.5, & \text{if } t_i \text{ is in the set of the factual core of the document,} \\ 1.0, & \text{in other cases.} \end{cases} \tag{6}$$

$$\lambda_{\text{ner}}(t_i) = \begin{cases} 1.5, & \text{if } t_i \text{ is part of a named entity;} \\ 1.0, & \text{in other cases.} \end{cases} \tag{7}$$

The term frequency is still computed as the sum of the weights of all its occurrences. This yields a new TF:

$$TF_{\text{new}}(t,d) = \sum_{t_i \in occ(t,d)} w_i \tag{8}$$

where t is the term, d the document, and $occ(t,d)$ the set of occurrences of the term t in the document d.

Finally, the final weight of a term in a document follows the TF-IDF structure:

$$W_{\text{new}}(t,d) = TF_{new}(t,d) \times IDF(t,D) \tag{9}$$

where D is the corpus of documents.

This method has been tested on large corpora of Russian news articles across specific domains: politics, finance, and sports. The authors systematically compare the classical TF-IDF with their new TF-SYN-NER-Rel method, using topic coherence as the main evaluation metric. They show that this new method improves both the coherence and stability of the generated topics while remaining computationally efficient.

In the context of term indispensability, this method can be used to extract topics. The score assigned by the method to these topics could then be interpreted as an indispensability score for a term within the specialized domain to which it belongs.

YAKE: Some other methods have further improved the statistical approach. Yet Another Keyphrase Extraction (YAKE) is a statistical approach that considers context and integrates lexical features like case, sentence position, term frequency, and dispersion. In Giarelis et al. [3], the method is described as splitting the text into individual terms and then assigning a score S(t) for each term t with the following metrics: $\mathrm{T_{case}}$, the casing aspect of a term; $\mathrm{T_{pos}}$ favors terms that are positioned near the beginning of the document; $\mathrm{TF_{norm}}$ term frequency normalization; $\mathrm{T_{rel}}$ term relatedness to context, which calculates the number of different terms that occur on the left and right side of the term; $\mathrm{T_{difsent}}$ which measures the frequency of a term in different sentences. The score is calculated as follows:

$$S(t) = \frac{T_{\text{rel}} \cdot T_{\text{pos}}}{T_{\text{case}} + \frac{TF_{\text{norm}}}{T_{\text{rel}}} + \frac{T_{\text{difsent}}}{T_{\text{rel}}}} \tag{10}$$

The YAKE method is not constrained by the source or collection of texts. It automatically identifies and extracts the most important keyphrases from the text by considering factors such as TF, position, and relevance. As said in Xu et al.'s survey [13], this allows it to be effective across various text collections and domains.

RAKE: Rapid Automatic Keyword Extraction (RAKE) is an unsupervised, domain-independent and language-independent algorithm based on word degree, frequency, and degree-to-frequency ratio for extracting keywords from individual documents. In Rose et al. [8], the score of a candidate keyword is computed as the sum of the scores of its constituent words, where each word is defined as the ratio between its degree (co-occurrence with other words) and its frequency in the text:

$$score(t) = \sum_{w \in t} \frac{d(w)}{f(w)} \tag{11}$$

Metric by Zhou and Salter: In reference [15], Zhou and Salter analyzed previous works that proposed using the gaps between successive occurrences to detect relevant words, assuming that important words appear in a non-uniform manner (in clusters). They propose a new metric based on the degree of word clustering, incorporating: spatial distribution, sequential order of occurrences and boundary conditions (beginning and end of text).

Their metric (Γ) measures the local density of words in specific areas and weights the occurrences according to their membership in clusters. This approach addresses several limitations of the previous methods. It is more stable in the face of local variations, capturing complex grouping structures more effectively and it operates on a single document without external references.

The proposed method represents a significant improvement over classical statistical approaches, introducing a more refined measure of relevance based on the structure of the text.

2.3 Embedding methods

Embedding-based methods work by representing terms as vectors, which means that words with similar meanings end up close to each other even if they look completely different on the surface, which is already an improvement on exact word matching. When it comes to indispensability, this is interesting because it can capture terms that are semantically central to a domain even if they are not frequent, which is exactly the kind of nuance that pure frequency-based methods like TF-IDF miss.

Dense vector embedding: Static embeddings such as Word2Vec and GloVe [6] were among the first methods to represent terms as dense vectors in a low-dimensional space. In order to extract keywords, these approaches compute the cosine similarity between candidate term embeddings and a document-level embedding, retaining the most similar terms.

To extend this idea, KeyBERT [3] replaces static embeddings with contextual representations produced by BERT [1], an encoder transformer model that generates context-dependent token representations for natural language understanding tasks, or its sentence-level variants such as Sentence_BERT. This allows

the model to account for polysemy and context-dependent meaning. Candidate keywords are ranked by their cosine similarity to the document embedding, under the assumption that the most representative keyphrases are those whose vector representation is closest to that of the document.

Dense embedding methods are characterized by expressiveness, but they don't produce explicit term weights, making it difficult to directly quantify the relative importance of individual terms within a domain. This limitation motivates the transition towards sparse learned representations and more sophisticated models such as SPLADE.

Sparse vector embedding: SPLADE [2] is a learned sparse retrieval model built on top of the BERT encoder architecture. The model produces sparse representations over the full vocabulary of the underlying language model, assigning a non-negative weight to each term in the vocabulary. SPLADE feeds the input text through a BERT encoder and applies the MLM (Masked Language Model) projection head to compute, for each vocabulary term j, a sparse weight aggregated over all input positions:

$$w_j = \sum_{i \in t} \log\left(1 + \mathrm{ReLU}(W_{ij})\right) \tag{12}$$

The ReLU ensures non-negativity, while the log-saturation prevents any single token from dominating. The result is a sparse vector $w \in R^{|V|}$ with typically only a few hundred non-zero dimensions out of $|V|$=30,522.

SPLADE is trained with a contrastive objective combining a ranking loss with FLOPS regularization to encourage sparsity:

$$\mathcal{L} = \mathcal{L}_{\text{rank}} + \lambda \cdot \mathcal{L}_{\text{FLOPS}} \tag{13}$$

A key property is implicit term expansion: the MLM head operates over the full vocabulary at each token position, allowing the model to activate terms semantically related to the input even if absent from it; for instance, the term "fracture" may activate "bone" or "break".

2.4 Graph-based approaches

Graph-based approaches convert documents into graphs where nodes represent terms and edges reflect how often they appear together, and from there centrality measures are used to determine which terms matter the most based on their position in the graph rather than just their frequency.

PageRank-based keyword ranking: A first family of graph-based approaches uses PageRank-style score propagation. The idea is simple: a document is converted into a graph where words are nodes and edges connect words that appear together in a sliding window, and a word is considered important if it is linked

to other important words. This is exactly what TextRank does. Zhang et al. [14] push this further with *TP-CoGlo-TextRank*, which adds signals like word frequency, position, co-occurrence frequency and GloVe-based similarity, making the propagation less uniform and a bit more nuanced.

The original TextRank score is defined as [14]:

$$S(V_i) = \frac{1-d}{N} + d \sum_{V_j \in In(V_i)} \frac{1}{|Out(V_j)|} S(V_j), \tag{14}$$

where $S(V_i)$ is the score of node V_i, N the number of words, $In(V_i)$ the set of nodes pointing to V_i, $Out(V_j)$ the outgoing neighbors of V_j, and d the damping factor. Zhang et al. [14] modify this by introducing a node weight and non-uniform transition probabilities:

$$S(V_i) = wt(V_i) \left[\frac{1-d}{N} + d \sum_{V_j \in In(V_i)} we(V_j, V_i)\, S(V_j) \right], \tag{15}$$

with $wt(V_i) = \lambda\, pos(V_i) + (1-\lambda)\, tf(V_i)$ and $we(V_j, V_i) = \gamma\, Co(V_j, V_i) + (1-\gamma)\, Glo(V_j, V_i)$, combining structural, statistical and semantic information within the same framework.

What is interesting in the experiments of Zhang et al. [14] is that not all signals contribute equally. Word frequency doesn't change much (TF-TextRank improves AF1 by only 0.47%), while word position has a much bigger impact (Pos-TextRank reaches 8.30% improvement), which is a significant gap for a single signal. GloVe similarity adds a little more on top of co-occurrence, though less than expected. Their final model, TP-CoGlo-TextRank, still achieves the best overall performance [14], even if position alone seems to carry most of the weight.

Algorithms	Top 7 (Uncombined)			Top 7 (Combined)		
	$AP\%$	$AR\%$	$AF_1\%$	$AP\%$	$AR\%$	$AF_1\%$
M1	37.21	53.32	43.83	43.16	54.03	47.99
M2	27.32	36.69	31.32	29.52	37.90	33.19
M3	38.44	54.93	45.23	45.50	55.65	50.06
M4	47.23	60.01	52.86	57.29	61.45	59.30
M5	38.66	54.86	45.35	45.60	55.67	50.14
M6	48.95	61.42	54.48	60.54	62.79	61.64

Table 1. Comparison of six keyword extraction algorithms reported by Zhang et al. [14]. TP-CoGlo-TextRank (M6) achieves the best performance in both settings.

This family of methods is relevant for indispensability because importance propagates through the graph rather than being a purely local property. A term

can score high not just because it appears a lot, but because it connects to other central or semantically related terms, which is much closer to how importance actually works in a specialized domain. The improved version of TextRank also shows that this mechanism can be enriched with position or semantic signals, making it a promising foundation for a more nuanced indispensability score.

That said, results depend heavily on how the graph is built, especially the window size and preprocessing choices. Some added signals such as GloVe similarity do not always bring the expected gains [14], which raises questions about added complexity. More importantly, the method targets keyword extraction, not domain-level term ranking, so its use for indispensability remains an indirect approximation.

Centrality-based Graph-of-Words weighting: A second approach works differently from PageRank: instead of propagating scores, the idea is to directly measure how central each term is in the graph. Malliaros and Skianis [4] represent documents as a *Graph-of-Words* (GoW) where nodes are terms and edges capture co-occurrence within a sliding window, and then use interpretable graph metrics like degree, in-degree, out-degree or closeness centrality to score each term, with no recursive propagation involved.

The simplest measure is degree centrality [4]:

$$degree(i) = \frac{|N(i)|}{|V| - 1}, \tag{16}$$

and closeness centrality:

$$closeness(i) = \frac{|V| - 1}{\sum_{j \in V} dist(i, j)}. \tag{17}$$

These weights can be further combined with IDF:

$$tw\text{-}idf(t, d) = tw(t, d) \cdot \frac{1}{1 - b + b \cdot |d|/avgdl} \cdot idf(t, D). \tag{18}$$

Results on Reuters-21578 R8 and WebKB are convincing. Degree centrality in undirected graphs consistently outperforms TF and TF-IDF baselines. On Reuters-21578 R8, TW-IDF degree reaches F1 0.9175 and accuracy 0.9661, and on WebKB the gap is more visible, with TW degree hitting F1 0.8962 against 0.8331 for TF-IDF [4], suggesting that graph structure captures something frequency alone tends to miss.

This approach captures the structural role of a term rather than just how often it appears. A term can be important because it co-occurs with many relevant terms or sits at a central position in the domain network, which is a very different kind of importance from TF-IDF. For vocabulary learning, this matters because the most indispensable terms are often the ones connecting several important notions, not just the most frequent ones.

Weighting	F1-score	Accuracy	Weighting	F1-score	Accuracy
TF	0.9127	0.9634	TF	0.8741	0.8853
TW, degree	0.8991	0.9611	TW, degree	**0.8962**	**0.9032**
TW, in-degree	0.8037	0.9438	TW, in-degree	0.8286	0.8545
TW, out-degree	0.8585	0.9546	TW, out-degree	0.8365	0.8603
TW, closeness	0.9125	0.9625	TW, closeness	0.8960	0.9004
TF-IDF	0.8962	0.9616	TF-IDF	0.8331	0.8538
TW-IDF, degree	**0.9175**	**0.9661**	TW-IDF, degree	0.8800	0.8882
TW-IDF, in-degree	0.8985	0.9629	TW-IDF, in-degree	0.7890	0.8381
TW-IDF, out-degree	0.8854	0.9625	TW-IDF, out-degree	0.8049	0.8474
TW-IDF, closeness	0.8846	0.9547	TW-IDF, closeness	0.8505	0.8674

Table 2. Comparison between classical term-weighting baselines and graph-based schemes on Reuters-21578 R8 and WebKB, adapted from Malliaros and Skianis [4].

Results depend on design choices like graph directionality, weighting, and window size, which are not always easy to justify. The method also stays local to the document and does not account for semantic similarity between different lexical forms. Finally, since the study [4] focuses on text categorization rather than vocabulary learning, its relevance to indispensability remains indirect.

2.5 Supervised methods

Supervised approaches to term importance are based on the idea of learning a scoring function from annotated corpora, where terms or phrases have been labeled according to their relevance within a specific domain, which means that the problem is usually formulated as a classification or ranking task, where models are trained to distinguish between terms that matter and terms that do not.

Early approaches in this direction relied on relatively simple architectures, which already showed promising results. Patra et al. [5], for instance, proposed a neural network combined with a relevance-based term weighting scheme, and showed that explicitly learning term importance could improve text classification performance in a meaningful way. Sheikh et al. [11] went in a similar direction by introducing a weighted Neural Bag-of-Words model that is capable of learning task-specific word importance, which made it possible to bridge the gap between more traditional weighting schemes like TF-IDF and more modern neural representations.

However, these methods face a fundamental limitation when it comes to modeling indispensability, which is that they all assume the existence of a stable ground truth established through human annotation, and that assumption is actually quite problematic since, as we discussed earlier, indispensability is an inherently subjective notion where two experts in the same field may simply not agree on whether a given term is important or not, making the construction of a reliable training dataset particularly difficult and expensive. As noted by

Ushio et al. [12], such annotated resources are costly to produce and often too context-specific to be reused in a different setting anyway.

On top of that, supervised models tend to generalize poorly from one domain to another, and a model that was trained on a given corpus will often struggle to adapt to a new domain without being retrained from scratch, which of course requires new labeled data, and so on, creating a kind of vicious circle that significantly limits how useful these approaches can be in dynamic or highly specialized contexts.

This is partly why recent literature has shown a growing interest in unsupervised or weakly supervised approaches that try to reduce this dependence on expert annotation, since Shang et al. [10] for instance pointed out that many term and phrase extraction methods still rely on complex linguistic analyzers or manually provided labels, which makes it very hard to extend them to new domains or languages without a significant amount of additional work.

2.6 Hybrid methods

Hybrid methods are basically about combining different approaches together to make up for what each one is missing on its own, mixing statistical, graph-based and semantic signals all at once.

For indispensability this makes a lot of sense, because no single method really captures everything on its own, statistical methods give you frequency and specificity, graph-based approaches tell you about structural centrality, and embeddings bring in semantic similarity, so putting them together gives a much more complete picture of how important a term actually is in a domain.

NBOW2: The paper by Sheikh et al. [11] proposes a significant improvement to Bag-of-Words models by incorporating a mechanism for learning word importance. The Bag-of-Words model is a model in which a document is represented as a vector of word frequencies. The methods used in this model are generally weighting schemes such as TF-IDF to identify important words. However, these approaches remain limited because they capture neither the semantics of words nor their specific importance for a given task.

With the emergence of deep learning methods, continuous vector representations of words, known as word embeddings, have been introduced. These representations, such as Word2Vec or GloVe, enable the capture of semantic and syntactic relationships between words. In this context, several neural models have been proposed for text classification, notably convolutional and recurrent neural networks.

Although these models achieve strong performance, they are computationally costly and require careful tuning of numerous parameters. In contrast, simpler architectures such as the Neural Bag-of-Words (NBOW) model offer an appealing alternative. It represents a document as the average of its word vectors and

uses this representation for classification. This model is fast and efficient, but it assumes that all words contribute equally to the final decision, which constitutes a significant limitation.

To address this issue, the authors propose an extension of NBOW called Neural Bag-of-Weighted-Words (NBOW2). This method introduces a word-weighting mechanism, allowing the model to automatically learn the importance of words based on the classification task. Specifically, the average of the word vectors is replaced by a weighted sum, where each weight is learned during model training. This weighting depends on the similarity between the word vector and a learned reference vector, which allows the capture of relevant semantic relationships.

Experiments conducted on sentiment and topic classification tasks show that the NBOW2 model improves performance compared to the traditional NBOW model. Furthermore, the weights learned by the model are interpretable and comparable to the weights obtained using traditional methods such as TF-IDF. This demonstrates that the model is capable of identifying the most discriminative words for a given task.

Despite its advantages, the method has certain limitations. It does not take into account word order in sentences and is less effective than more complex models such as convolutional or recurrent neural networks.

SIFRank: SIFRank and SIFRank+ [3] are unsupervised methods which combine statistical and embedding-based approaches with two different contributions: the ELMo pre-trained language model and the SIF sentence embedding model.

ELMo (Embeddings from Language Models) is a deep contextualized word representation model based on a bidirectional LSTM architecture trained on a large-scale corpus. It produces dynamic, context-dependent representations for each token, unlike static word embedding models (Word2Vec or GloVe): the same word receives a different embedding vector depending on the surrounding context.

SIF (Smooth Inverse Frequency) handles the problem of aggregating token-level embeddings into a single sentence-level vector. A naive average of word embeddings is dominated by high-frequency function words ("the", "of", "and"), which contribute little semantic content. SIF mitigates this by weighting each token according to:

$$\mathrm{w}(t) = \frac{a}{a + p(t)} \tag{19}$$

where a=10^{-3} is a smoothing parameter and p(t) is the relative frequency of the token t in the corpus. Rare tokens receive weights close to 1 while frequent function words receive weight close to 0. Then the sentence embedding is computed as the weighted average of the token embeddings.

These two components are integrated by SIFRank with cosine similarity between each candidate embedding and the document embedding, and the top-n candidates are returned ranked by descending score:

$$\text{SIFRank}(v_{ck_i}, v_d) = \text{CosSim}(v_{ck_i}, v_d) = \frac{v_{ck_i} \cdot v_d}{\|v_{ck_i}\| \cdot \|v_d\|} \tag{20}$$

SIFRank+ extends this approach to handle longer documents, where keyphrases appearing earlier in the text tend to be more representative of the document's main topic. It multiplies the cosine similarity score by a position-biased weight derived from a modified inverse frequency function:

$$\text{SIFRank+}(v_{ck_i}, v_d) = \frac{e^{\text{Pos}(ck_i)}}{\sum_{j=1}^{N} e^{\text{Pos}(ck_j)}} \cdot \text{SIFRank}(v_{ck_i}, v_d) \tag{21}$$

where $Pos(ck_i)$ is defined as:

$$\text{Pos}(ck_i) = \frac{1}{p_l + \mu} \tag{22}$$

with p_l denoting the position of the keyphrase's first occurrence in the document and μ an offset parameter. This formulation allows the model to assign higher scores to keyphrases appearing closer to the beginning of the document, given that earlier-occurring terms tend to carry greater topical relevance. SIFRank is recommended for short documents such as abstracts, while SIFRank+ is better suited for full-length papers.

3 Limitations of existing approaches

Existing approaches for computing the indispensability index often struggle with polysemy and contextual ambiguity, and their performance tends to be highly domain-dependent, limiting generalization across different topics, languages, and datasets. In addition, obtaining high-quality data is costly, as previously discussed, particularly due to the difficulty of defining the notion of indispensability. In this section, we review the limitations of each type of method.

Term extraction methods are a necessary step, but one that remains imperfect. They still present several important limitations that can affect their overall performance. A major issue lies in their reliance on frequency-based signals, which often leads to overlooking rare but meaningful domain-specific terms. Moreover, pattern-based approaches may struggle with linguistic variability, failing to capture expressions that do not conform to predefined syntactic structures. Even more advanced systems such as AutoPhrase rely on external corpora or pre-trained models, which can introduce bias or reduce their effectiveness when applied to highly specialized or low-resource domains. In addition, segmentation processes are not always accurate and may incorrectly merge or split terms, thereby reducing semantic coherence.

Statistical methods are simple and efficient, but are limited in that they rely on shallow distributional information. They capture how often and where terms occur, but not what they represent in terms of meaning, which limits their ability to model semantics in a strong way. These methods regard words as stand-alone units, and so fail to capture linguistic facts such as synonymy, polysemy or semantic equivalence. Hence, two terms that mean the same thing can get very different scores depending on how they are distributed in the corpus. Besides, they are also very sensitive to the shape and volume of the corpus they are run on. IDF-based methods, for instance, can blow up when they come across small or very homogeneous datasets where the statistics on global frequency are not that trustworthy. Last but not least, some may be based too much on superficial patterns like word co-occurrence or positional distribution to the extent that these features become a proxy for terms requiring conceptual depth, particularly in complex or noisy texts.

Embedding-based methods have achieved significant success, but are also not devoid of limitations. First, while the representations are sparse, they still rely on a large fixed vocabulary which can hold them back when facing unseen or domain-specific terms. In addition, the implicit expansion feature can negatively affect precision by activating a bunch of terms that have nothing to do with the topic at hand. Training the whole joint ranking and sparsity objective is also a relatively slow and computationally expensive task. And while the sparsity constraint does the method a favor in terms of speed, it makes it less informative than dense embeddings. There is also the question of its use within the framework of the BERT architecture, which brings its own set of limitations, biases, and challenges related to context. Ultimately, one might argue that it is somewhat interpretable. However, translating a non-zero weight into a truly meaningful interpretation remains a considerable semantic leap.

Graph-based approaches, although effective in certain contexts, also present several structural limitations. First, they rely heavily on graph construction choices (window size, edge weighting, term filtering), which can significantly affect the results without any universal rule for optimal tuning. Second, these methods mainly capture local co-occurrence relationships, which limits their ability to model deeper semantic links or implicit conceptual relations between terms. Moreover, the frequent lack of normalization across documents makes it difficult to directly compare scores on the scale of a full corpus. PageRank-based approaches also introduce non-negligible computational complexity, particularly for large graphs, while simpler centrality measures may lack precision in assessing importance. Finally, these models remain sensitive to lexical noise (frequent but uninformative terms, morphological variations), especially in the absence of advanced linguistic processing.

Despite advances in supervised learning approaches, several fundamental limitations persist and hinder their large-scale deployment. First, the reliance on annotated data introduces bias stemming from the annotators' choices, which can undermine the consistency and reproducibility of the results. Second, these

models often require large amounts of data to perform effectively, which becomes problematic in fields where such resources are scarce or costly to acquire. Furthermore, the complexity of certain architectures can lead to overfitting, which reduces their ability to function well in new contexts. Finally, the lack of interpretability in certain models makes it difficult to understand the criteria that lead them to judge a term as important.

Finally, hybrid approaches constitute an attempt to strike a balance between several types of methods, but they are not without limitations. First, they heavily rely on the quality of pre-trained embeddings, which can introduce biases stemming from training corpora and limit their ability to generalize to specialized domains. Moreover, these methods do not explicitly account for syntactic or discourse dependencies between words. Their unsupervised nature may also limit their adaptation to specific tasks requiring a fine-grained understanding of context. Finally, although less computationally expensive than complex deep learning models, they still depend on computational resources for contextual embedding computation. These limitations suggest that there is still a trade-off to be found between simplicity, interpretability, and performance.

4 Potential future directions

As discussed earlier in this paper, indispensability is an inherently subjective notion; therefore, one of the main challenges that researchers will have to address is how to evaluate it in a way that is both practical and reproducible, especially since relying solely on direct human annotation tends to be costly, time-consuming, and often inconsistent between annotators. Therefore, two complementary directions for evaluation should be explored. On one hand, large language models could be used not just for embedding generation, but also as actual evaluators of term indispensability through prompting strategies, where, for example, a model could be asked to rate how essential a given term is for understanding a domain or to compare the relative importance of several terms at once, which would provide scalable and weakly supervised signals that could approximate human judgment while still being adaptable across different domains. On the other hand, proxy indicators derived from real-world usage could also serve as practical approximations, since terms that appear frequently in introductory sections of scientific papers or in educational materials like textbooks and course syllabi are likely to reflect what the community considers fundamental concepts.

Another direction that seems most promising is the design of hybrid pipelines that explicitly combine several complementary signals into a single framework, for example by using statistical methods like TF-IDF to capture term specificity, graph-based approaches to model structural centrality within the domain, and contextual embeddings to encode semantic relevance, and then integrating all of these signals into a unified scoring function that would produce more robust and interpretable estimates of indispensability than any single method alone.

Of course, none of this would be possible without reliable data, and progress in this area will therefore also depend on the availability of large and high-quality domain-specific corpora, since resources like large academic databases could be leveraged to build coherent datasets that cover multiple domains, languages, and levels of specialization.

One last direction that has not yet received much attention in the existing literature is the question of how term indispensability evolves over time, because the importance of a term within a domain is probably not something fixed and stable, but rather something that changes as the field itself changes, as new methods appear, and as certain older concepts become less central or even obsolete. In domains like medicine or computer science for instance, terms that were once considered absolutely fundamental can progressively lose their importance when new techniques replace older paradigms, while on the contrary some emerging concepts can gain prominence very quickly and become indispensable within just a few years. This suggests that indispensability should probably be modeled as a dynamic quantity that evolves together with the domain, rather than as a single fixed score that is computed once and never updated.

5 Conclusion

In this paper, we looked at the problem of estimating the indispensability of a term within a specialized domain, which proves to be more challenging than initially expected. To tackle this, we reviewed several families of existing approaches, from statistical and embedding-based methods to graph-based, supervised and hybrid techniques. Each captures something useful, whether that is frequency, semantic similarity, or structural centrality, but none of them fully works on its own for something as subjective and domain-dependent as indispensability.

What our analysis really shows is that combining these complementary signals is the most promising direction, and future work should focus on building and evaluating such combined approaches on specialized corpora, along with better evaluation frameworks for this inherently subjective problem. In the long run, this could really help with domain-specific vocabulary learning and concept prioritization.

References

1. Devlin, J., Chang, M.W., Lee, K., Toutanova, K.: Bert: Pre-training of deep bidirectional transformers for language understanding. In: Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers). pp. 4171–4186 (2019)
2. Formal, T., Piwowarski, B., Clinchant, S.: Splade: Sparse lexical and expansion model for first stage ranking. In: Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval. p.

2288–2292. SIGIR '21, Association for Computing Machinery, New York, NY, USA (2021). https://doi.org/10.1145/3404835.3463098, https://doi.org/10.1145/3404835.3463098

3. Giarelis, N., Karacapilidis, N.: Deep learning and embeddings-based approaches for keyphrase extraction: a literature review. Knowledge and Information Systems **66**(11), 6493–6526 (Nov 2024). https://doi.org/10.1007/s10115-024-02164-w, https://doi.org/10.1007/s10115-024-02164-w
4. Malliaros, F.D., Skianis, K.: Graph-Based Term Weighting for Text Categorization. In: Proceedings of the 2015 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining 2015. pp. 1473–1479. ACM, Paris France (Aug 2015). https://doi.org/10.1145/2808797.2808872, https://dl.acm.org/doi/10.1145/2808797.2808872
5. Patra, A., Singh, D.: Neural Network Approach for Text Classification using Relevance Factor as Term Weighing Method. International Journal of Computer Applications **68**(17), 37–41 (Apr 2013), https://ijcaonline.org/archives/volume68/number17/11674-7301/
6. Pennington, J., Socher, R., Manning, C.: GloVe: Global vectors for word representation. In: Moschitti, A., Pang, B., Daelemans, W. (eds.) Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP). pp. 1532–1543. Association for Computational Linguistics, Doha, Qatar (Oct 2014). https://doi.org/10.3115/v1/D14-1162, https://aclanthology.org/D14-1162/
7. Rodionov, D., Konnikov, E., Golikov, G., Yakob, P.: Structural–Semantic Term Weighting for Interpretable Topic Modeling with Higher Coherence and Lower Token Overlap. Information **17**(1) (Dec 2025). https://doi.org/10.3390/info17010022, https://www.mdpi.com/2078-2489/17/1/22
8. Rose, S., Cowley, E, W., Crow, V., Cramer, N.: Rapid automatic keyword extraction for information retrieval and analysis (01 2009), https://www.osti.gov/biblio/1039881
9. Salton, G., Yang, C.S., Yu, C.T.: A theory of term importance in automatic text analysis. Journal of the American Society for Information Science **26**(1), 33–44 (1975). https://doi.org/https://doi.org/10.1002/asi.4630260106, https://asistdl.onlinelibrary.wiley.com/doi/abs/10.1002/asi.4630260106, _eprint: https://asistdl.onlinelibrary.wiley.com/doi/pdf/10.1002/asi.4630260106
10. Shang, J., Liu, J., Jiang, M., Ren, X., Voss, C.R., Han, J.: Automated Phrase Mining from Massive Text Corpora (Mar 2017). https://doi.org/10.48550/arXiv.1702.04457, http://arxiv.org/abs/1702.04457, arXiv:1702.04457 [cs]
11. Sheikh, I., Illina, I., Fohr, D., Linarès, G.: Learning Word Importance with the Neural Bag-of-Words Model. In: Proceedings of the 1st Workshop on Representation Learning for NLP. pp. 222–229. Association for Computational Linguistics, Berlin, Germany (2016). https://doi.org/10.18653/v1/W16-1626, http://aclweb.org/anthology/W16-1626
12. Ushio, A., Liberatore, F., Camacho-Collados, J.: Back to the Basics: A Quantitative Analysis of Statistical and Graph-Based Term Weighting Schemes for Keyword Extraction. In: Moens, M.F., Huang, X., Specia, L., Yih, S.W.t. (eds.) Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing. pp. 8089–8103. Association for Computational Linguistics, Online and Punta Cana, Dominican Republic (Nov 2021). https://doi.org/10.18653/v1/2021.emnlp-main.638, https://aclanthology.org/2021.emnlp-main.638/

13. Xu, K., Feng, Y., Li, Q., Dong, Z., Wei, J.: Survey on terminology extraction from texts. Journal of Big Data **12**(1), 29 (Feb 2025). https://doi.org/10.1186/s40537-025-01077-x, https://doi.org/10.1186/s40537-025-01077-x
14. Zhang, L., Li, Y., Li, Q.: A Graph-Based Keyword Extraction Method for Academic Literature Knowledge Graph Construction. Mathematics **12**(9) (Apr 2024). https://doi.org/10.3390/math12091349, https://www.mdpi.com/2227-7390/12/9/1349
15. Zhou, H., Slater, G.W.: A metric to search for relevant words. Physica A: Statistical Mechanics and its Applications **329**(1), 309–327 (Nov 2003). https://doi.org/10.1016/S0378-4371(03)00625-3, https://www.sciencedirect.com/science/article/pii/S0378437103006253